Communicating Your Work





written reports

“I’m going to tell you about a problem and how I solved it.”

"I'm going to tell you about a problem and how I solved it in much more detail."

story elements



motivation
description
approach
learning
result

how is writing different?

presentation

“Two people were injured today, and traffic was tied up for two hours, after a driver lost control of a tractor-trailer truck on the Zakim Bridge.”

written report

“A driver lost control of a tractor-trailer truck on the Zakim Bridge today, causing an accident that injured two people and tied up traffic for two hours.”

how is writing different?

presentation

"Two people were injured today, and traffic was tied up for two hours, after a driver lost control of a tractor-trailer truck on the Zakim Bridge."



A speaker often presents the "news" first, followed by the context.

how is writing different?

written report

"A driver lost control of a tractor-trailer truck on the Zakim Bridge today, causing an accident that injured two people and tied up traffic for two hours."

Writing is generally more effective when you present the context first. The "news" should be toward the end of the sentence, in the "stress position."

how is writing different?

ba-da-BUM

how is writing different?

- Backward-linking old information appears in the topic (first) position.

  "In order to determine which algorithm to use,...

- The subject (person, thing or concept the story is about) also appears in the topic position.

  "our team...

- The new, emphasis-worthy information appears in the stress position.

  "... first conducted a literature search."

how is writing different?

In order to determine which algorithm to use, our team first conducted a literature search. We found...



flow: end of sentence introduces what follows

not

Our team first conducted a literature search in order to determine which algorithm to use. We found...

some simple rules

- Provide context before asking the reader to consider anything new.
- Put your actions into verbs.
- Omit unnecessary words.
- Use the stress position at the end of the sentence to give your sentences impact.

some simple rules

- Follow a subject with a verb as soon as possible.

DNA in repeat regions or with long stretches of the same base or small microsatellites causes problems for next-gen sequencers.

DNA causes problems for next-gen sequencers whenever it is in repeat regions. Long stretches of the same base or small microsatellites also cause problems.

some simple rules

- Put your actions into verbs.
- Omit unnecessary words.

implied action

Detailed analyses of the evolutionary features of different types of regulatory elements are an important area for future research.

explicit action

Future research should analyze the evolutionary features of different types of regulatory elements.

beware of "nominalizations"

- actions that appear in parts of a sentence other than verbs

implied action

Our analysis using regression and k-means clustering revealed...

explicit action

We analyzed the data with regression and k-means clustering. This analysis revealed...

a "good" nominalization summarizes the action of the previous sentence



use active voice when possible

- Rewording in the active voice frequently sharpens the text.
- Varying active and passive can be effective.

The substrate surface was mapped using an Atomic Force Microscope.

The AFM mapped the substrate surface.

There were five mice receiving antibiotics.

Five mice received antibiotics.

We have introduced a method to support fast rotation invariant search of large shape datasets with arbitrary representations and distance functions. Our method supports rotation limited queries and mirror image invariance if desired.

Future work includes both extensions and applications of the current work. We will attempt to extend this approach to the indexing of 3D shapes, and we have begun to use our algorithm as a subroutine in several data mining algorithms which attempt to cluster, classify and discover motifs in a variety of anthropological datasets, including petroglyph and projectile point databases.

The VLDB Journal (2009) 18:611–630
Supporting exact indexing of arbitrarily rotated shapes and periodic time series under Euclidean and warping distance measures

Eamonn Keogh · LiWei · Xiaopeng Xi · Michail Vlachos · Sang-Hee Lee · Pavlos Protopapas

We plan to extend and apply this approach, beginning with the indexing of 3D shapes. In addition, we have begun to use our algorithm as a subroutine in several data mining algorithms used in anthropology to cluster, classify and discover motifs in a variety of datasets, including petroglyphs and projectile points

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- active voice
- reduced nominalizations

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- inserted transition phrase
- removed repeated phrase "attempt to"

In addition, we have begun to use our algorithm as a subroutine in several data mining algorithms used in anthropology to cluster, classify and discover motifs in a variety of datasets, including petroglyphs and projectile points.

thanks to:

Duke Scientific Writing Resource

http://cgi.duke.edu/web/sciwriting/index.php

parting thoughts

- use words for what words do best: **express**
  - connect with the audience
    - build trust, show authority
  - convey nuance, sentiment, humor, emphasis
  - create flow and rhythm
  - explain graphics
    - interpret, place in context
    - label information

parting thoughts

- use pictures for what pictures do best: **show**
  - reveal patterns
  - cut through complexity
  - reinforce learning
  - display methodology
  - provide visual relief and pleasure
  - add visual humor

parting thoughts

- use symbols for what symbols do best: **elaborate**
  - reveal mathematical structure of your work
  - aid understanding for technical readers and nonverbal learners
  - display methodology
  - increase information density

the goal

understanding, enjoyment and appreciation of your work by **all** readers and listeners



it's worth the effort!

thanks for listening
...questions?